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Venue B / Room 312

Structural Analysis of Regional Economic Circulation and Interregional Leakage

An Input-Output Approach to the Shikoku Region (1985-2005)

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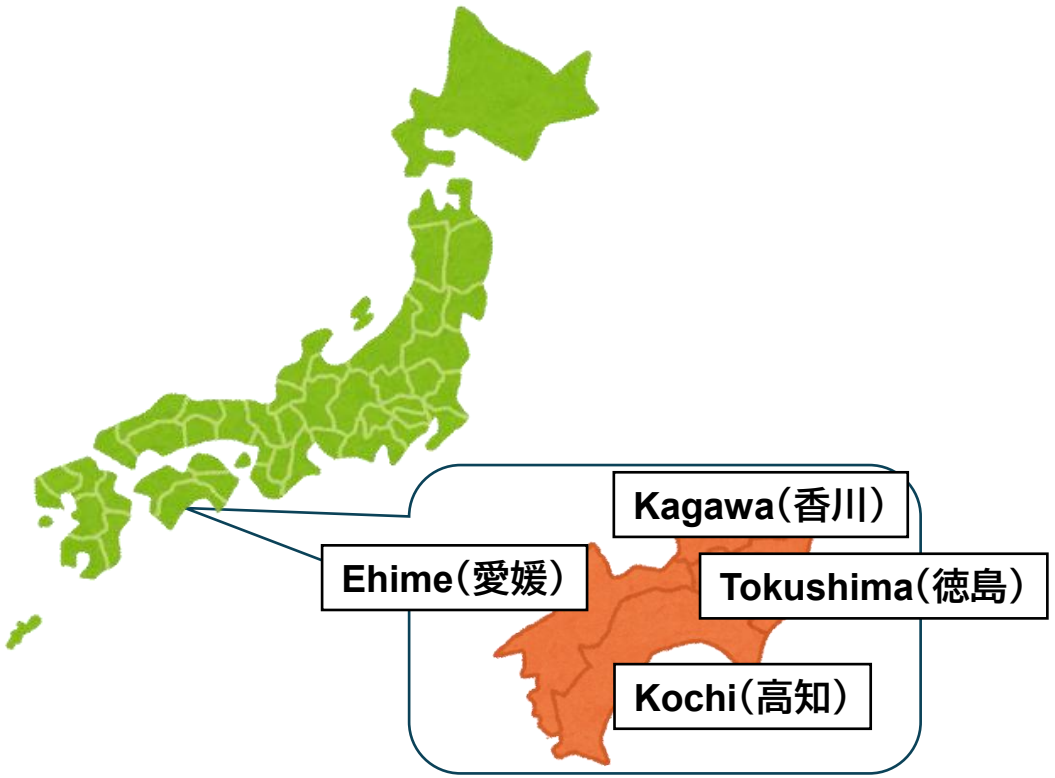
What is Shikoku（四国）?

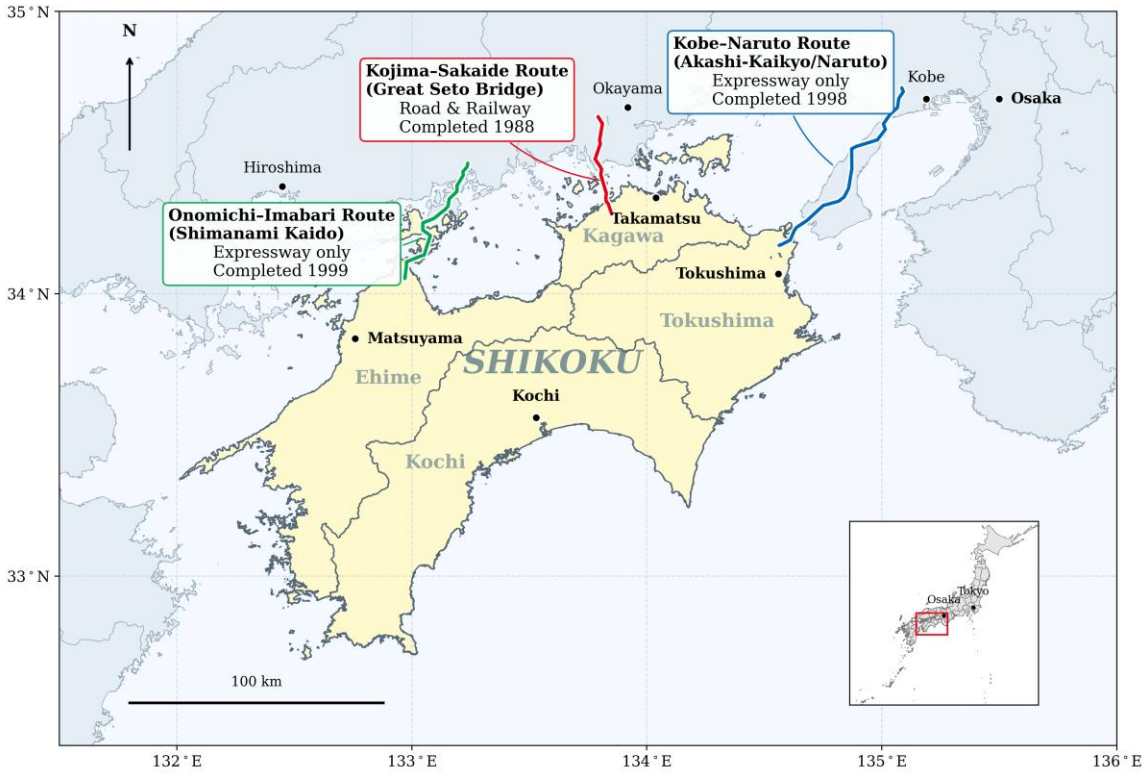


Population	**3.6 million people (2025)** Approximately 3% of Japan's total population
GDP	**¥14.4 trillion (2015) / $91.8 billion** ($1 = ¥ 157) Approximately 3% of Japan's total GDP.
Land Area	**18,803 ㎢** Approximately 5% of Japan’s total land area The smallest of Japan's four main islands.
Industry	**Distinct Industrial Belts** *North (Seto Inland Sea):* Heavy Industry (Shipbuilding, Chemicals, Paper). *South (Pacific):* Agriculture, Forestry, Fishery.

The Connectivity Paradox

Completion of the Honshu-Shikoku Bridges (1985–1999)

Q: Does better infrastructure boost the regional economy — or accelerate the drain to Tokyo?



Key Concept

Straw Effect (Economic drainage to the core via infrastructure) suggested by Ono in 1980s.

Source: Prepared by the authors based on the Ministry of Land, Infrastructure, Transport and Tourism(MLIT) National Land Numerical Information (Administrative Boundaries Data).

Research Objectives: Shikoku as a Structural Laboratory

Why analyze Shikoku?

Shikoku serves as an ideal **“laboratory”** for examining center-periphery dynamics.
Shikoku is a compact economic unit (3% of national GDP) that serves as a demographic early case. It mirrors Japan’s national challenges—depopulation and aging—**approximately 20 years in advance.**

1. Structural Shift in Economic Autonomy

Investigating the transition from **autonomous economic circulation** to **structural dependency** on external trade.

2. Identification of Structural Drivers

Determining if self-sufficiency decline is driven by the **Share Effect** (service-shift) or the **Rate Effect** (loss of competitiveness).

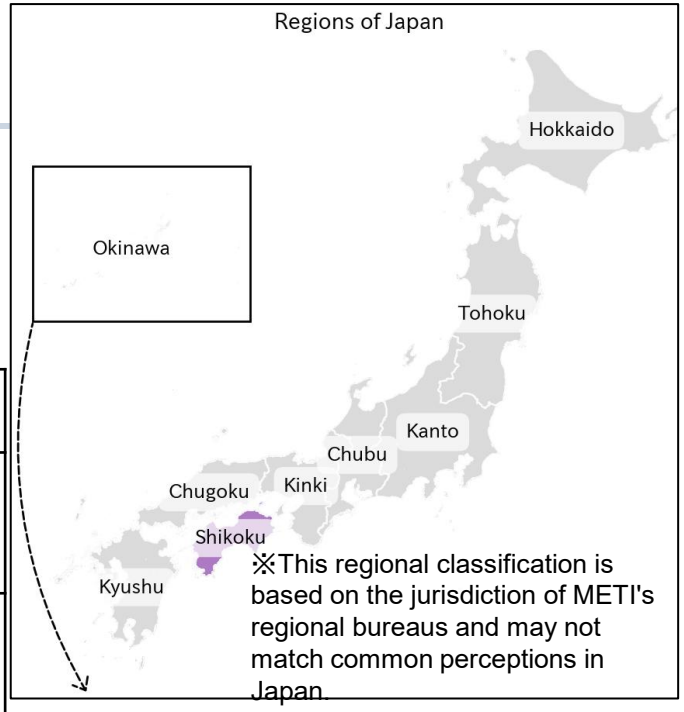
3. Mapping Value Leakage Destinations

Mapping value leakage to assess the development of a **"branch economy"** versus **global market integration**.

Data Sources and Classifications

Interregional Input-Output Tables

Published by	the Ministry of Economy, Trade and Industry (METI)
Coverage	9 regions, 35 sectors, 4 benchmark years (1985, 1990, 1995, 2005)
Sectors	Original classifications (45–53 sectors) were harmonized to ensure consistency over time.
Prices	Current prices.
Note	The 2005 table is the final edition, as METI later discontinued compilation.



Methodological note: We focus on "exchange value" (a regional purchasing-power footprint) rather than physical volume. (Discussed later.)

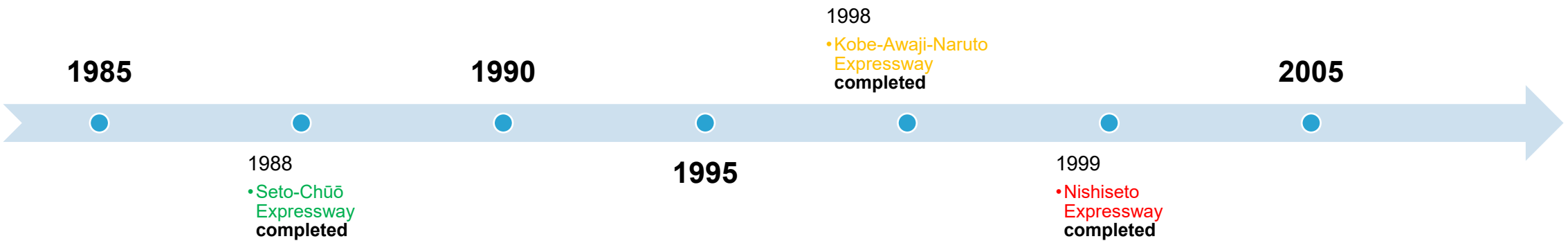
Why focus on 1985–2005?

1. **Data continuity**
   METI's nine-region Interregional Input–Output Tables (IRIOT) are consistently available only up to 2005, making 1985–2005 the longest comparable window.

2. **Two structural shocks**
   The period covers
   (i) the 1988 Great Seto Bridge opening, shifting Shikoku from maritime separation to land-based integration with Honshu
   (ii) post–Plaza Accord yen appreciation, which accelerated globalization and manufacturing relocation.

3. **Demographic relevance**
   Shikoku's population peaked in 1985—over two decades before Japan's 2008 peak—so it serves as an early case of structural adjustment under depopulation.

→**This window uniquely combines the best long-run IRIO data with key shocks and early demographic decline.**



Methodology

- This study employs IO framework-based visualization to intuitively extract structural shifts in regional economies
  - The methodological framework for the skyline and balance charts is primarily based on Miyagawa (2005) and Uda (2019), both of which were published in the official journal of PAPAIOS.
- Analysis Environment: Python 3.11.13 (via Google Colaboratory)

Previous Research (Interregional IO Tables)

Prior IO studies point to centralization and weak internal linkages—our study revisits this with a skyline approach.

1. Hitomi et al. (2000)
   1980–1990, local intermediation declines and interregional trade expands, reinforcing Kanto's dominance.
   (Based on METI IRIO Table)
2. Suhara, Guo & Ishiro (2020)
   In 2010, Shikoku is largely self-sufficient in aggregate, but inter-prefectural linkages are weak.
   (Based on the authors' estimated table)

Gap: We still lack a sector-by-sector view of how self-sufficiency can persist while the export base narrows —especially relative to Kanto.

Review of Our Previous Study (PAPAIOS Conference 2025 Autumn) and Current Issues

Average Propagation Length (APL) quantifies the average number of steps involved in the propagation of economic impacts, derived from the interregional Leontief inverse.

Previous Analysis

PAPAIOS Autumn Conference (Oct 2025)

We applied **Average Propagation Length (APL)** analysis to the same 1985-2005 harmonized dataset.

Finding

Shikoku's supply chains shortened significantly compared to the national average.

Interpretation

Efficient global integration, but also a sign of "structural thinning" (loss of intermediate stages).

The Missing Link

APL measures "how long" the chain is, but not **"where" the value flows**.

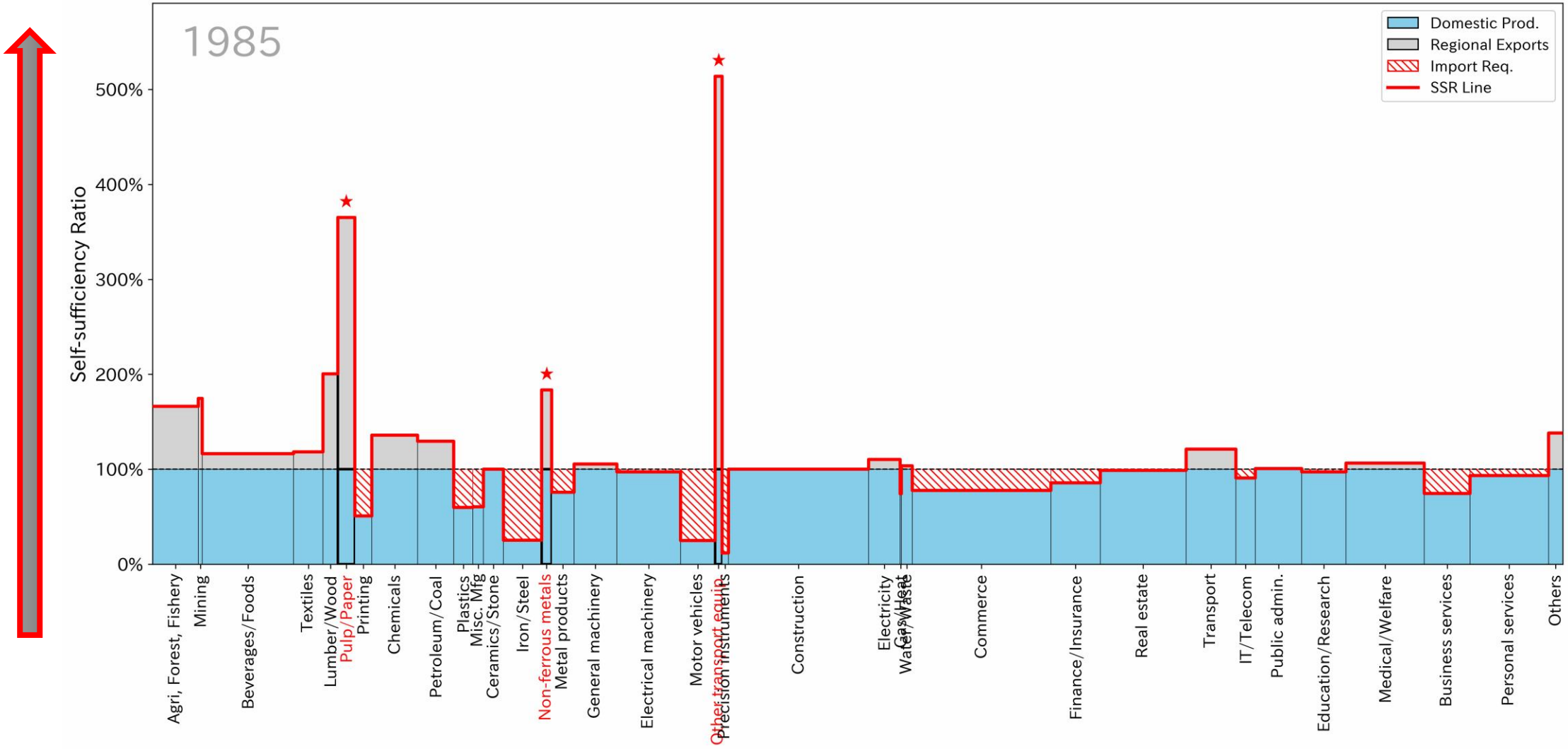
This Study's Focus

We employ **Skyline & Leakage Analysis** to decompose the destination of value.

Key Question
Is the value leaking to Tokyo or Overseas?

Skyline Charts (Shikoku)

Limited sectors rise



Service sectors expand

Skyline Charts: What We Learn

Services expand, while export capacity concentrates into a few narrow export niches ("thin needles").

Structural vulnerability behind service expansion
Service growth can reflect a deepening dual structure, not broad diversification.
Employment shifts locally, but foreign-currency earning capacity becomes concentrated in a narrowing export base.

Asymmetric export stability within a shrinking base
Some matureprocess industries act as stable surplus anchors across cycles.
In contrast, project-based sectors can generate large activity but add high volatility and exposure to global shocks.

Two shifts in the skyline charts
(i) **Intensive competitiveness increases** (SSR ↑)
(ii) **Extensive scale contracts** (width ↓)

$$\text{(i) Intensive Competitiveness: } \frac{\partial SSR_i}{\partial t} > 0$$

$$\text{(ii) Extensive Scale: } \frac{\partial w_i}{\partial t} < 0$$

Theoretical Framework: From NEG to Structural Sharpening

We define "Structural Sharpening" as a condition where specific industries appear to gain competitiveness only because they are shedding marginal capacity.

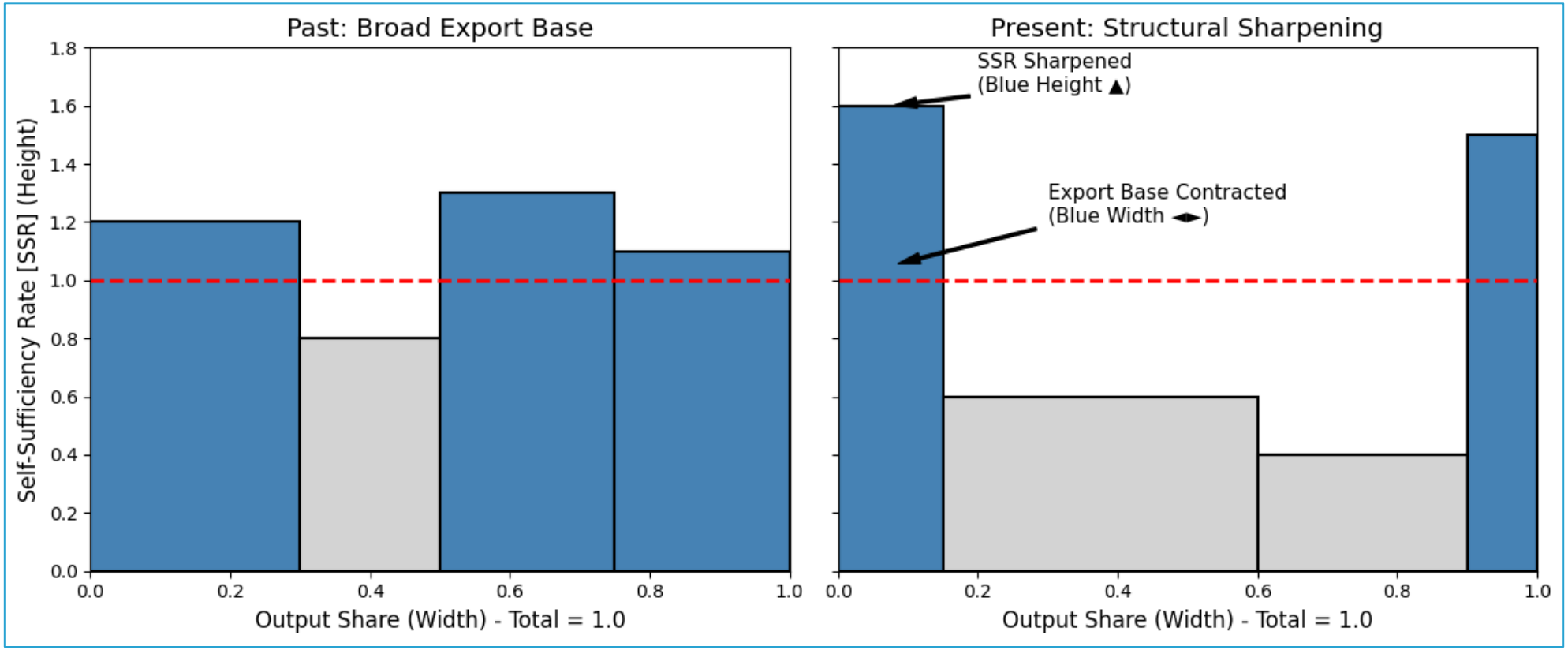
Dynamics	Standard NEG Theory (New Economic Geography)	**Concept: Structural Sharpening**
Industrial Relations	Complete Agglomeration Core absorbs all industrial activity.	**Strategic Persistence** Periphery retains specific niches.
Height = SSR	Self-sufficiency ratio (SSR) declines as dependency grows.	⤴**Maintained or Increased** (Self-sufficiency Rate)
Width = Output share	Functional specialization remains stable.	⤵**Significantly Contracted** (Output Share)
Core Logic	Functional shift (R&D to Core).	**Survival strategy** based on scale contraction.

Key prediction: Height (SSR) holds up or rises, while Width (output share) contracts.

Theoretical Framework: From NEG to Structural Sharpening

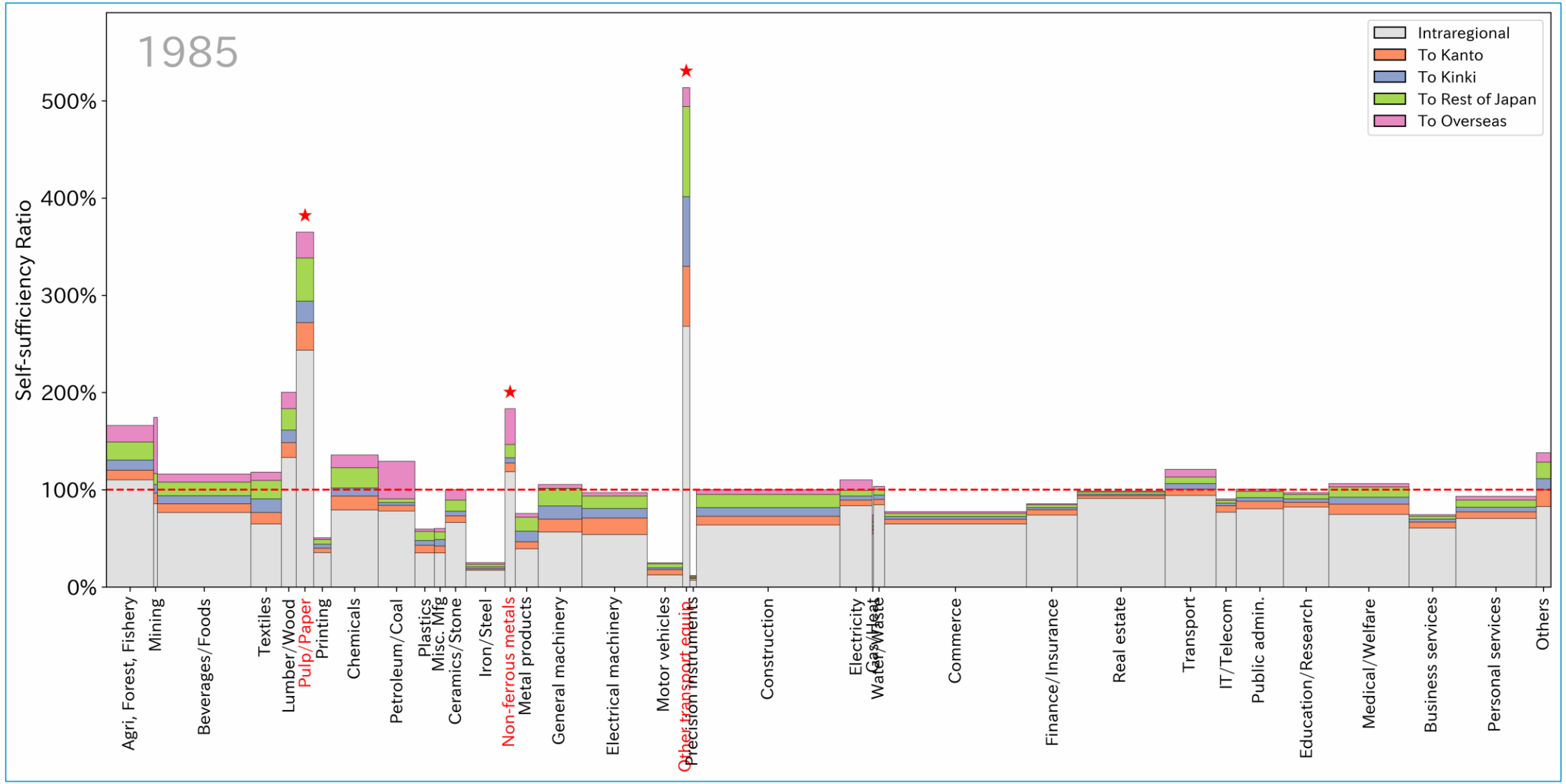
We define **"Structural Sharpening"** as a condition where specific industries appear to gain competitiveness only because they are shedding marginal capacity.

Competitiveness can improve mechanically through contraction: narrower base, higher SSR = **stronger but smaller.**

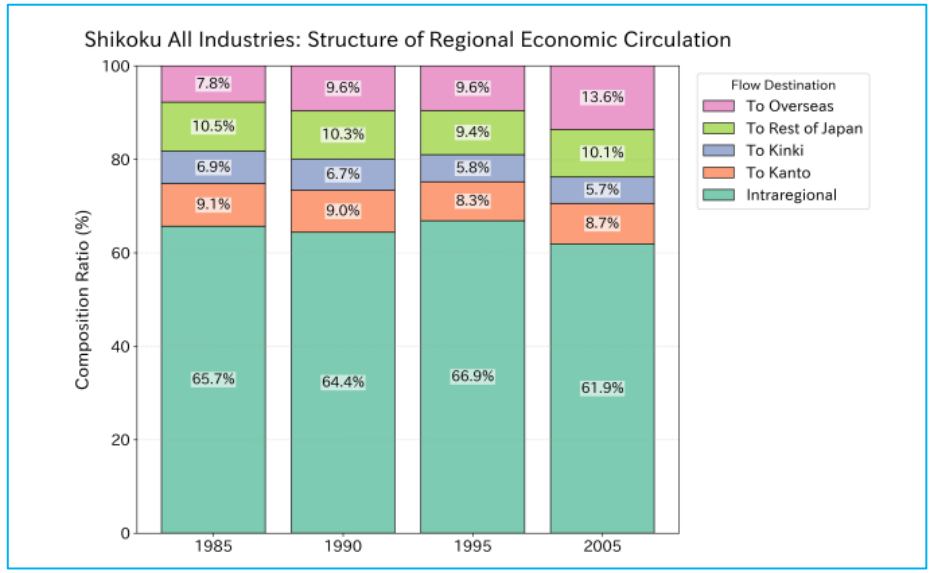


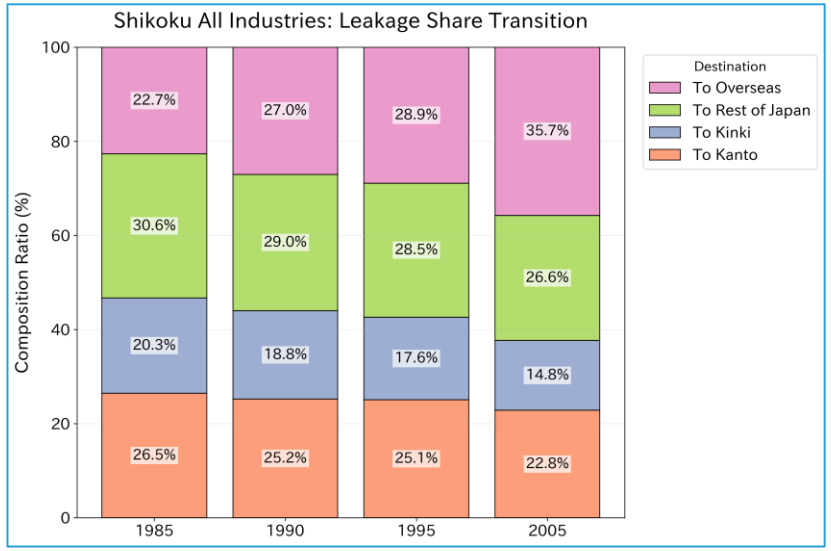
SSR can rise when marginal segments exit and the remaining base becomes more selective.

Decomposition of Leakage : Balance Charts (Shikoku)



The Dual Leakage Structure





1. Rapid Integration into Global Markets

Rapidly increased in overseas leakage (22.7% → 35.7%)

Driven by offshoring of supply chains

2. Weakening Domestic Interdependence

Declining economic ripple effects to Kinki and other regions

3. Persistent Reliance on Tokyo (Kanto)

Leakage remained rigid at approximately 9%

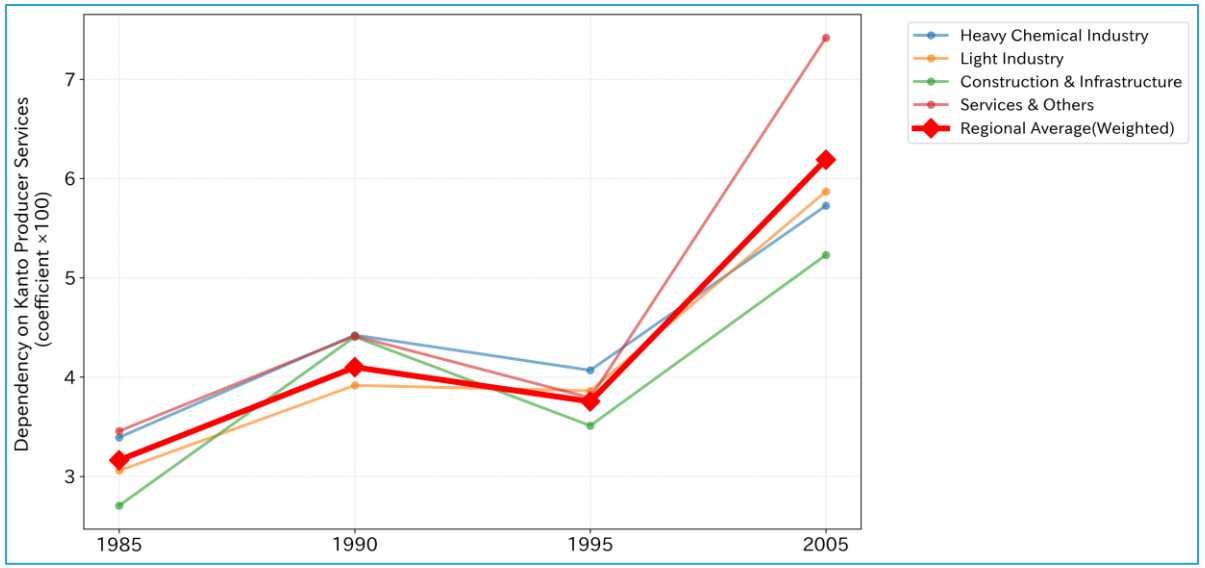
Continued dependence for professional/business functions

Dual leakage channels (Overseas & Tokyo) limit intraregional multiplier effects despite export growth.

Functional Dependence: The "Branch Economy

Brains in Tokyo, Factories in Shikoku

Kanto producer-service dependence index (IO-based inducement) in Shikoku



For each Shikoku sector, we calculate the total output induced in **Kanto's Finance, ICT, and Business Services** by **1 unit of final demand**
(Leontief inverse; coefficient × 100).

Core (Tokyo): "The Brains"

High value-added functions concentrate in the core: headquarters coordination, R&D, marketing, finance, and IT.
Agglomeration economies allow the core to capture producer-service income.

Periphery (Shikoku): "The Factories"

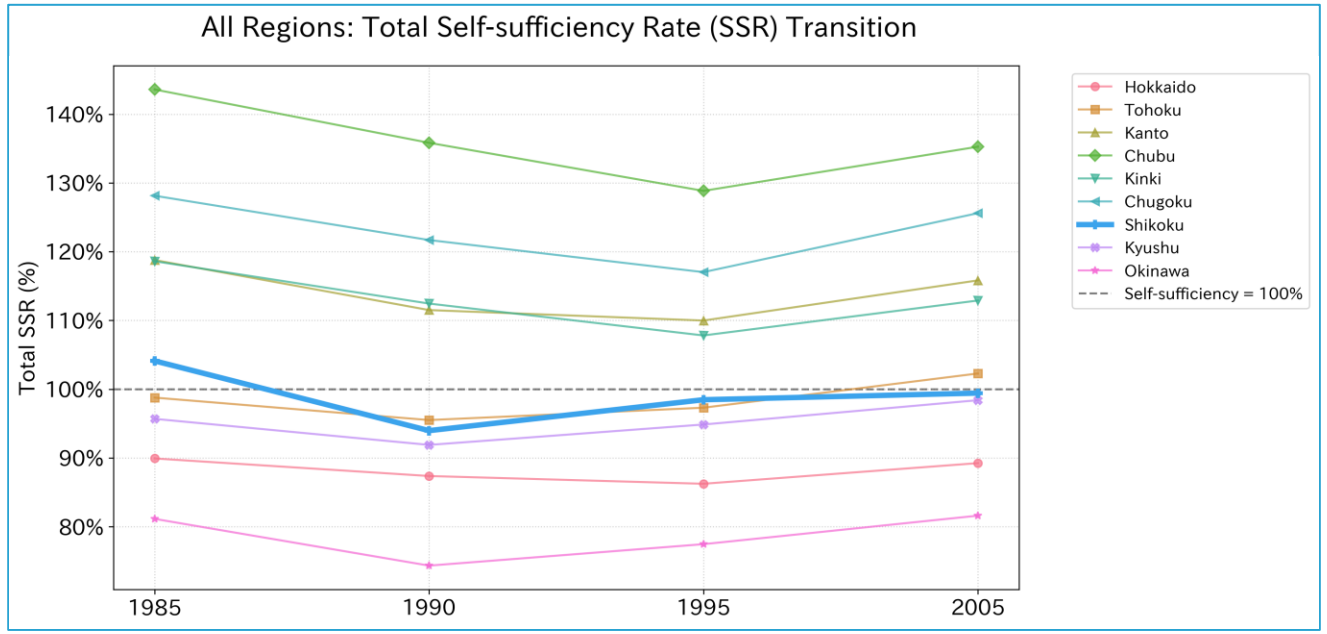
Capital-intensive production stages remain in the periphery.
Sunk costs and resource/port advantages make plants costly to relocate.

→This explains why export growth may not translate into stronger local multipliers when the producer-service layer is increasingly induced in Tokyo.

Did the Bridges Affect Shikoku's Supply Chain?

Shikoku experienced the sharpest SSR drop in Japan in 1985–1990

- Shikoku SSR drops **1.041** → **0.940** (1985–1990)
- The decline is larger than the national average decline
- Timing overlaps with the **1988 bridge opening** and major macro shocks (e.g., yen appreciation)



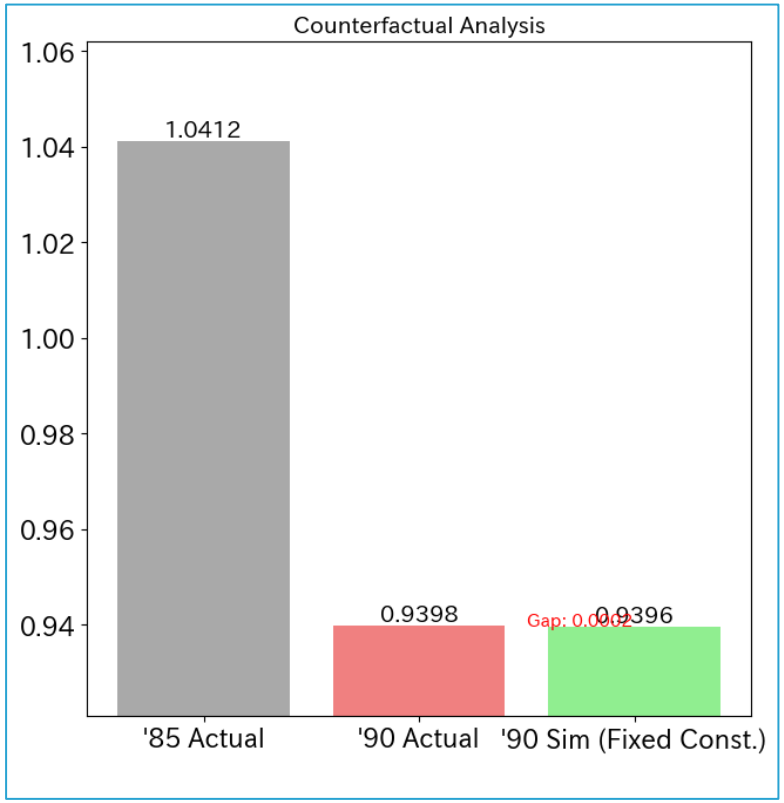
Did the Bridges Affect Shikoku's Supply Chain?

Analysis of the 'straw effect' and structural leakage via SSR decrease decomposition (1985–1990)

Did the bridge construction cause the economic decline in Shikoku?
What if Shikoku's construction demand stayed at 1985 levels?

- Simulated SSR (0.9396) ~Actual SSR (0.9398)
- Difference is only 0.0002 (Negligible).

→**Construction was NOT the driver of SSR decline.**



•1. Fix Share: Revert 1990 construction share to 1985 levels.
•2. Adjust Others: Pro-rate other sectors to ensure a 100% total.
•3. Simulate SSR: Derive hypothetical SSR using adjusted weights and 1990 data.

Did the Bridges Affect Shikoku's Supply Chain?

Analysis of the 'straw effect' and structural leakage via SSR decrease decomposition based on Dietzenbacher & Los (1998)

$$\Delta S = S^{t+1} - S^t = \underbrace{\sum_i (w_i^{t+1} - w_i^t) \cdot \bar{s}_i}_{\text{Share Effect}} + \underbrace{\sum_i \bar{w}_i \cdot (s_i^{t+1} - s_i^t)}_{\text{Rate Effect}}$$

What people buy changed
Measures the impact of changes in industrial composition while keeping sector-specific self-sufficiency rates fixed at 1985 levels

Where people buy from changed
Measures the impact of changes in regional self-sufficiency within each sector while keeping industrial shares fixed at 1985 levels

Note: Symmetric decomposition used. Interaction is negligible and redistributed.

Did the Bridges Affect Shikoku's Supply Chain?

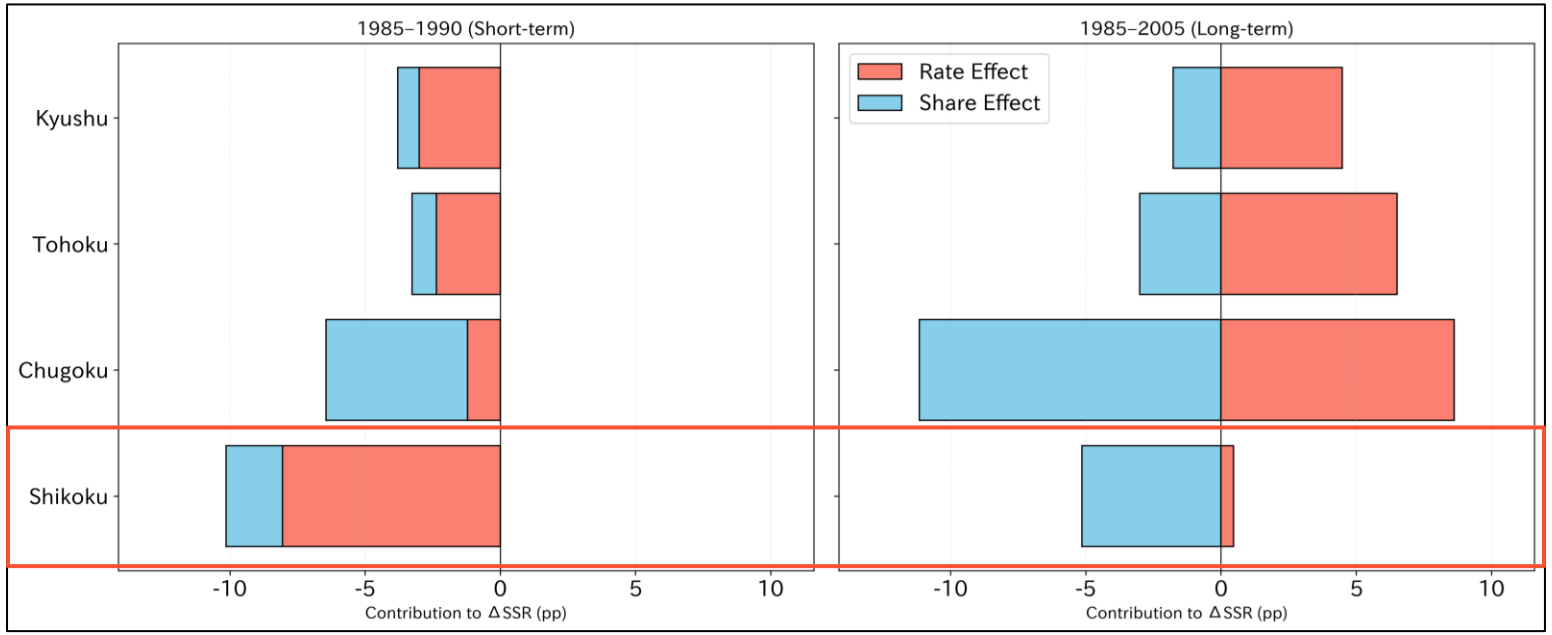
Bridges coincide with a two-stage change: short-run Rate collapse, long-run Share-driven decline

Short-run shock (1985–1990)
Rate-dominant drop
SSR −10.1 pp
(largest among peripheral regions)

Long-run shift (1985–2005)
Share-dominant decline
No offsetting positive Rate effect
(unlike Chugoku/Kyushu)

Rate effect
= "where we buy from changed"（within-sector sourcing/competitiveness）

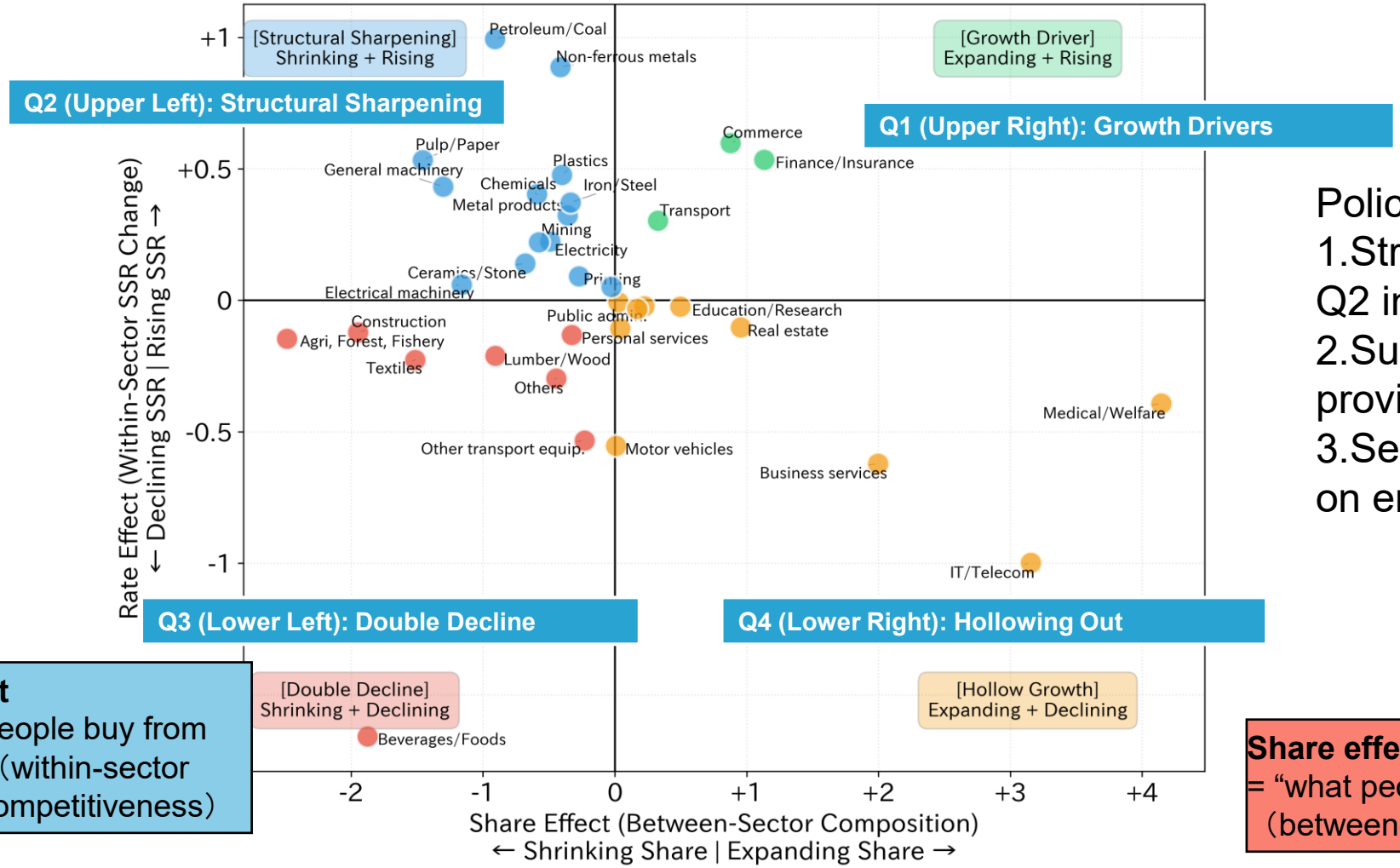
Share effect
= "what we buy changed"（between-sector composition）



Not a price artifact:
Real Share Effect
−7.89 pp (stronger)

Did the Bridges Affect Shikoku's Supply Chain?

SSR decrease decomposition on a scatter plot (1985-2005)



Policy Implications

1. Strengthen backward linkages of Q2 industries (thicken thin spires)
2. Substitute Q4 services with local providers
3. Selective support for Q3 based on employment impact

Rate effect
= "where people buy from changed" (within-sector sourcing/competitiveness)

Share effect
= "what people buy changed" (between-sector composition)

Additional: Robustness Check (Constant-Price Adjustment)

The result is not driven by relative prices. If anything, the negative share effect is larger in real terms.
Background: manufacturing prices fell relative to services over this period.

- **Question:** Could relative price changes (Baumol) inflate the share effect?
- **Test:** Recompute 1985–2005 decomposition using constant-price demand shares (2005=1).
- **Answer:** No. The share effect becomes more negative; the main conclusion is unchanged.

Region	Normal(Current)		Real(Constant 2005)	
	Share Effect	**Rate Effect**	**Share Effect**	**Rate Effect**
Shikoku	-5.1 pp	+0.5 pp	-7.9 pp	+3.5 pp
Chugoku	-11.2 pp	+8.6 pp	-10.3 pp	+10.3 pp
Kyushu	-1.8 pp	+4.5 pp	-2.7 pp	+4.6 pp

Shikoku: Share effect **-5.1** → **-7.9** pp; Rate effect **+0.5** → **+3.5** pp
Comparison: Chugoku and Kyushu show strong real rate effects that offset tertiarization.

Our main analysis uses nominal IO tables because we focus on regional exchange value and purchasing power.
The constant-price adjustment is therefore a robustness check, not the primary metric.

Measurement of headquarters-related services: what is (and isn't) captured?

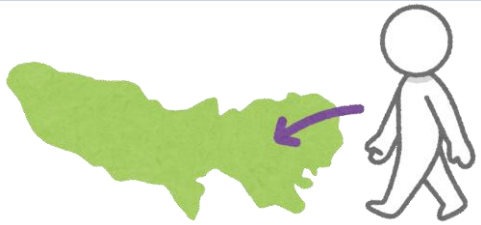
Two measurement caveats (levels are uncertain)
Caveat 1
IRIOT service flows are partly constructed.
Not fully observed bilateral transactions; imputed and balanced for macro-consistency (METI 2010).
Particularly relevant for producer services and HQ-type intra-firm coordination.

Caveat 2
HQ-related flows are sensitive to allocation/balancing rules.
Estimated magnitudes can change materially (Arai & Kim 2017; Suhara 2021).

How to interpret our dependency coefficient
Best seen as a harmonized structural indicator
Use for relative comparisons across sectors and periods
Not a point estimate of exact HQ service purchase volumes

Interpret coefficients as harmonized indicators (comparative), not levels.

Conclusion - The Reality of Economic Integration



The Structural Shift of Shikoku's Economy (1985-2005)

- **Structural Sharpening**
  - The regional export base contracted significantly from 41.0% to 32.5%. The economy evolved from a stable, diversified base into a highly vulnerable structure reliant on a few "needle-like" industries.
- **Functional Subordination**
  - Despite retaining physical production facilities, strategic functions such as R&D, finance, and management have systematically centralized in Tokyo.
- **The "Straw Effect" and Dual Economy**
  - Improved physical connectivity accelerated the replacement of local supply chains with external ones. This resulted in a polarized economy split between competitive global niche sectors and low-productivity local services.

Policy Implications - Strategic Re-bundling

Policy Implications for Peripheral Regions

•**Strategic Re-bundling:** Regional policy must pivot from simply promoting export volume to actively rebuilding internal economic circulation.

•**Thickening the Spires:** It is critical to foster local backward linkages to stabilize highly specialized but isolated industries, transforming fragile "needles" back into stable industrial bases.

•**Functional Autonomy:** Regions must develop local IT and professional business services to substitute for the high-order service imports currently flowing from the capital.

•**Sequencing Infrastructure:** Large-scale connectivity projects must be paired with local capability upgrades *before* opening to prevent competitive collapse.

Fractal Warning for Japan

Shikoku‘s structural sharpening foreshadows Japan’s future: risk of becoming a "Branch Economy" for global service hubs

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